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INFO 640 - Data Analysis
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Final Product Report

## Research Focus

Following my initial exploration of the relationship between higher documented audit rates for recipients of the Earned Income Tax Credit (EITC) and its correlation to their race or ethnicity, I decided to change the focus of my research question. I was interested in extending existing efforts to investigate the issue[3,4] but was not able to access granular enough information regarding individual taxpayers to replicate previous efforts[4]. Rather than keeping a narrow focus, I expanded my investigation and decided to look at the relationships between profiles of taxpayers such as their identified racial/ethnic category, their occupation (if they are currently employed), their level of reported income, and their likelihood of being at risk of an audit. Since tax collection and enforcement policies are often shaped by political ideology[2] and special interest group lobbying, I will then look at the voting record for the counties to see if there is a particular partisan lean to those counties with the highest audit rates.

## Redeveloped Research Question

Comparing available data from IRS records, United States Census surveys, and voter records from 2012 to 2016, are counties with higher audit rates more likely to have higher proportions of individuals identifying as Black, Hispanic, Asian American, Hawaiian-Pacific Islander, or Native American; employment concentration within specific industries; lower median household income; and demonstrate partisan voting patterns?

*$H_0$ (Null Hypothesis)* - Counties with higher observed audit rates from 2012 to 2016 will not be associated with higher proportions of racial and ethnic minority residents, higher levels of employment in certain industries, lower median income household levels, and higher levels of support for a particular political party.

*$H_1$ (Alternative Hypothesis)* - Counties with higher observed audit rates from 2012 to 2016 will be associated with higher proportions of racial and ethnic minority residents, higher levels of employment in certain industries, lower median income household levels, and higher levels of support for a particular political party.

## Data Set-up

To shape my model, I began with Bloomquist's dataset from 2019 which drew from IRS data observed from 2012 to 2015[3]. As part of his calculations, Bloomquist developed an "audit rate" as a means to normalize the observations across counties within the United States to account for the wide variation in populations. This constructed variable provides an outcome variable that reflects the intensity of enforcement actions taken against individuals within certain geographical areas, which is defined at the county level. As the IRS does not directly collect

certain demographic identifiers for respondents (importantly that of race and ethnicity), I looked to Census data to provide different demographic and sociological counts to associate with the county level data for the audit rate including race/ethnicity reported by household, household income levels, and the number of individuals employed in a given industry at the county level. To normalize the data and account for potential wide differences across the populations of different counties, I used percentages of individuals employed within a specific industry as well as the percentage of individuals with a certain household income which I divided into three broad categories (Low, Middle, and High Income).

Finally, to incorporate partisan lean into the analysis, I ingested a dataset developed by a team of researchers from University of Michigan's Institute for Social Research[7]. These researchers took data on voter registration and turnout over the years from the Election Administration and Voting Survey datasets which they combined with presidential and senate elections counts at the county level across the continental United States and Hawaii beginning in 2004. With this electoral data and voter turnout numbers, they constructed "partisanship indices" as a means to both reflect the partisanship within the county and normalize for voting population sizes.

Within the attached R script, I import Bloomquists' dataset providing estimated audit rates at a county level from 2012 through 2015; pull in specific demographic variables from the 5-year Annual Census Survey in 2015 including household compositions of race, income levels, and employment transforming those variables to percentages to allow for analysis; and integrate the constructed partisanship indices for 2016 which draw from electoral data from 2010 to 2016. After importing and normalizing the respective datasets, I merged them all using the common county level "FIPS" code.

## Initial Model Considerations

Before diving into analysis, I first wanted to look at the potential collinearity of my variables as the way in which I previously structured my data (percentages and indices totalling to 100%) would very likely cause collinearity issues. To remove that possibility, I decided to drop a few variables from my four sets of demographic and sociological factors. Ultimately, I chose to drop the variable reflecting the percentage of white households at the county level, my "Middle Income" and "High Income" categories which consisted of household incomes above $50,000 a year, the percentage of the employed individuals working in the Public Administration and the "Other" sectors, and then the Partisanship Index reflecting previous Democratic voting trends from registered voters. To help select the variables to exclude, I referred to existing literature which has identified trends in tax enforcement actions including a tendency to target Black, lower income (specifically those receiving Earned Income Tax Credit benefits) taxpayers at a higher rate[4]. After this initial reorganization and exclusion, I moved on to step-wise regression to further identify predictor variables to potentially exclude from my final model.

## Step-wise Regression

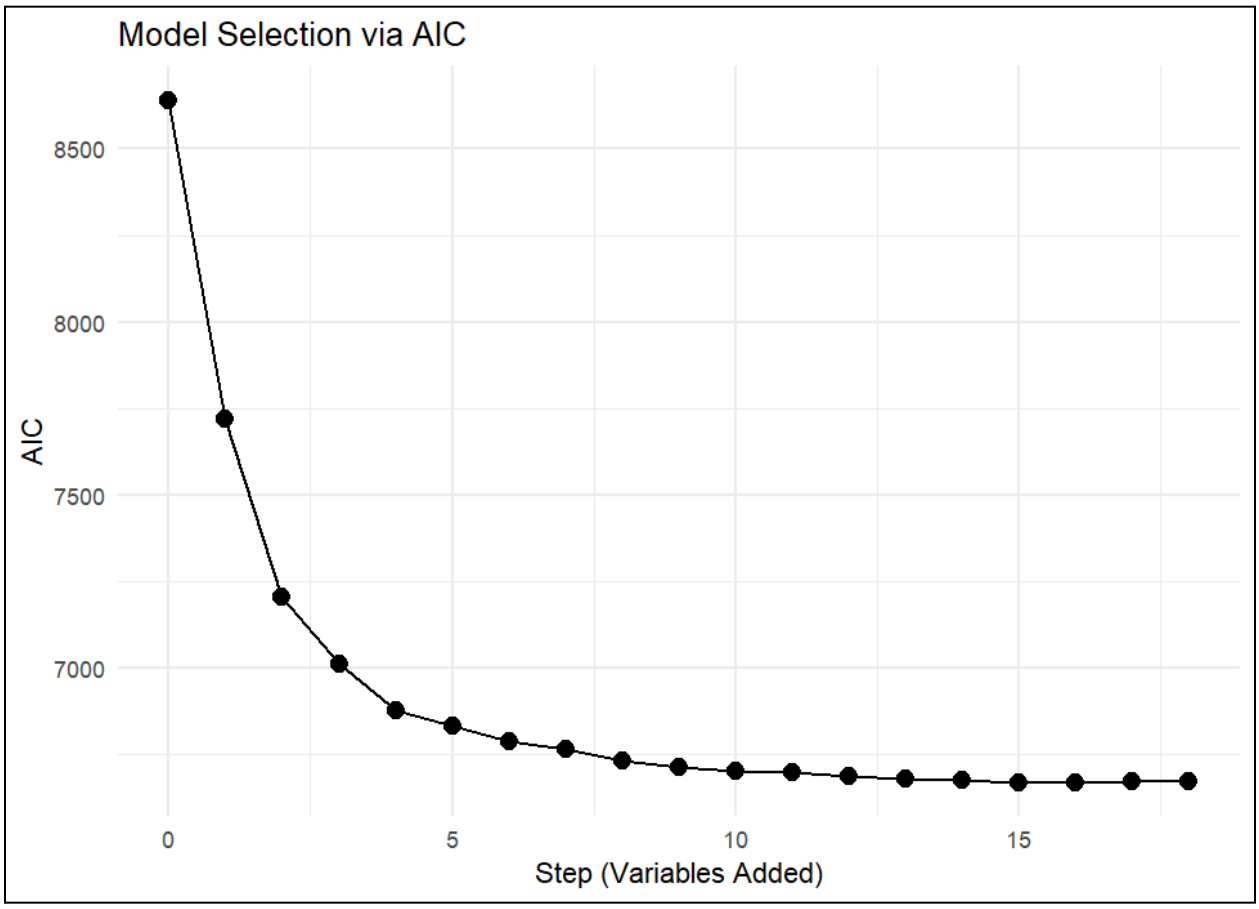
Observing that all of my variables are continuous, a multiple linear regression seemed like the best course for analysis. Before proceeding with that analysis, I ran a stepwise selection process on my model to see how the individual variables contribute and determine if there are ones that should be excluded for future analyses. To do so, I first created a linear regression model data object, “model”, containing all of my predictor variables compared against “audit_rate” as my outcome, using the “lm()” function. Next, I used the “ols_step_both_r2()” function from the “olsrr” library to run the step-wise regression test and then input the newly created data object “all_variables_both” to see the resulting values. Included in the results is a column of the r-squared values as the test added additional variables and continued its calculations. To help inform my interpretation, I also plotted the resulting forward stepwise modelling process which is reproduced below:

| **Stepwise Summary** | | | | |
|---|---|---|---|---|
| **Step** | **Variable** | **AIC** | **Adj. R2** | **R Value Increase** |
| 0 | Base Model (no predictors) | 8640.127 | 0.00000 | N/A |
| 1 | BlackPercent | 7718.049 | 0.25681 | 0.25681 |
| 2 | LowerIncome | 7206.312 | 0.36977 | 0.11296 |
| 3 | HispanicPercent | 7011.875 | 0.40816 | 0.03839 |
| 4 | NativeAmPercent | 6878.024 | 0.43327 | 0.02511 |
| 5 | Agri_MiningPerc | 6834.857 | 0.44126 | 0.00799 |
| 6 | PARTISAN_INDEX_REP | 6790.556 | 0.44934 | 0.00808 |
| 7 | ManufacturingPerc | 6766.591 | 0.45375 | 0.00441 |
| 8 | Arts_Rec_FoodPerc | 6733.921 | 0.45963 | 0.00588 |
| 9 | AsianPercent | 6714.832 | 0.46311 | 0.00348 |
| 10 | ConstructionPerc | 6701.435 | .46559 | 0.00248 |
| 11 | WholesalePerc | 6689.995 | 0.46772 | 0.00213 |
| 12 | Education_HealthPerc | 6680.046 | 0.46959 | 0.00187 |
| 13 | RetailPerc | 6676.161 | 0.47042 | 0.00083 |
| 14 | Finance_RealEstatePerc | 6671.177 | 0.47144 | 0.00102 |
| 15 | Science_WastePerc | 6670.781 | 0.47167 | 0.00023 |

| Stepwise Summary | | | | |
|---|---|---|---|---|
| 16 | TransportationPerc | 6672.150 | 0.47161 | -0.00006 |
| 17 | Hawaiian_PIPercent | 6673.736 | 0.47151 | -0.0001 |

As previously discussed in class, the AIC is somewhat limited as an interpretative tool when not used to compare the AIC values of different models, however, we can glean information from the values returned from the step-wise regression testing. As variables were added, the AIC value decreased at each step which reflects a better error predication ability of the model with each addition. We see a similar increase in the amount of variation explained by the model as the adjusted r squared value increases as well with the addition of more of our predicator variables.

The plot below more clearly demonstrates the decrease in AIC values as more variables are added to the model. It also highlights the specific importance of the first five variables and points to the decreased benefit of the addition of variables after the ninth step.

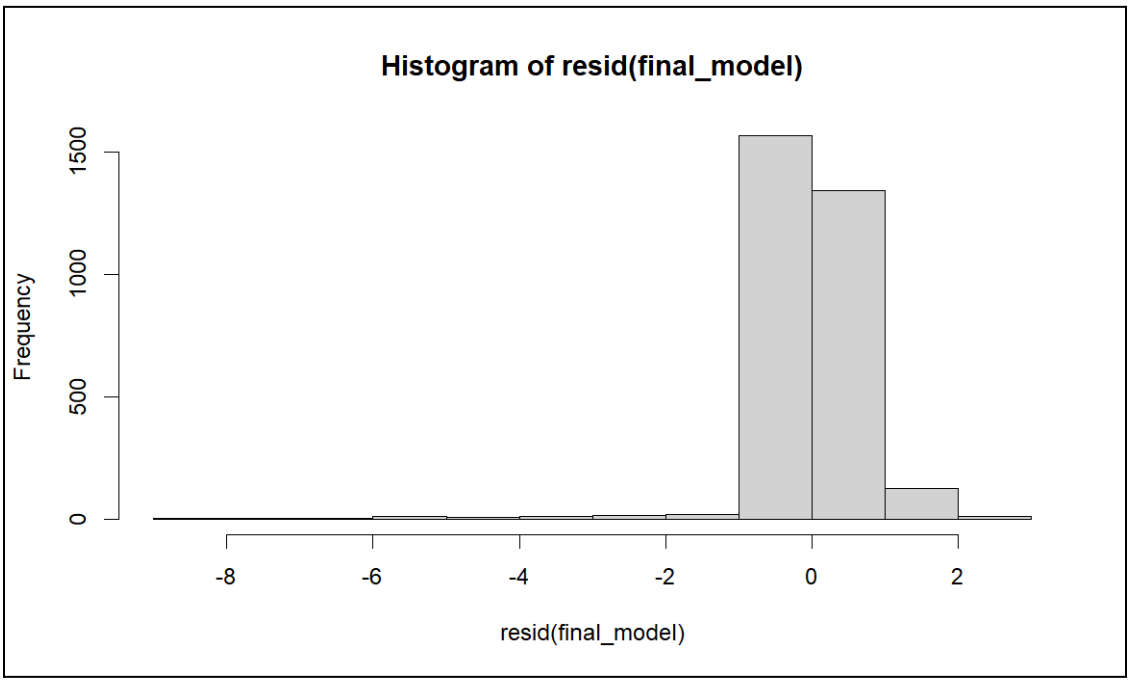


Considering that that each additional step after the ninth only adds .0025 or less to the overall adjusted r-squared value of the model, I decided to exclude those eight variables: “ConstructionPerc,” “WholesalePerc,” “Education_HealthPerc,” “RetailPerc,” “Finance_RealEstatePerc,” “Science_WastePerc,” “TransportationPerc,” and “Hawaiian_PIPercent.” Within my R script, I then filtered out those predictor variables, creating a new linear regression model containing only those remaining predictors which impacted the fit of the model in a more significant way.

## Testing Assumptions

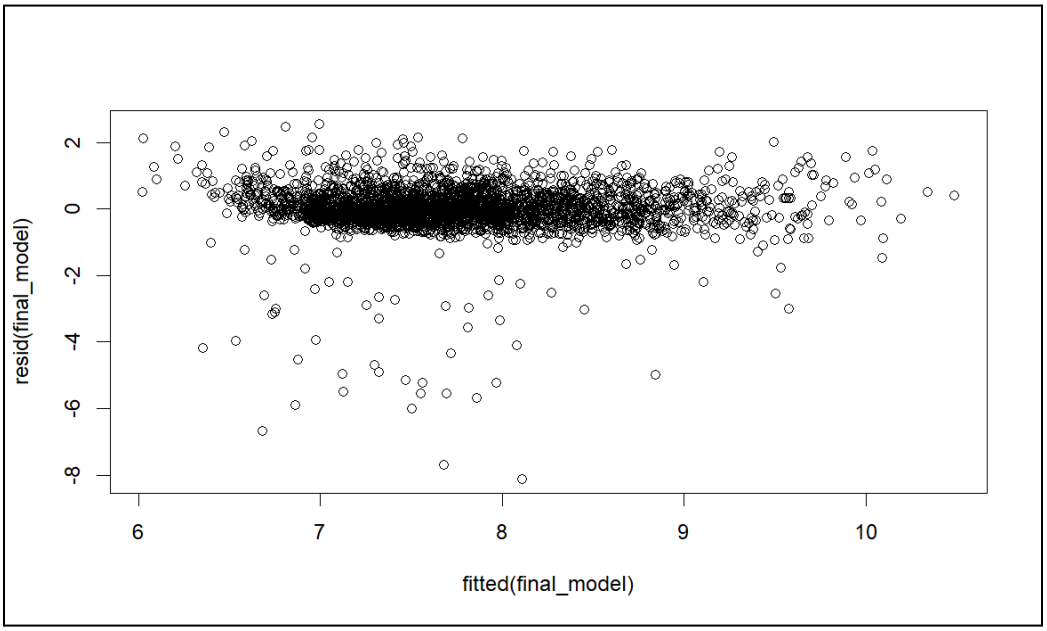
With this new model established, I ran through a few tests to check assumptions, evaluating the suitability for linear regression analysis and assessing the reliability of the results.

*Assumption 1*: Normal Distribution of Residuals: “hist(resid(model))”



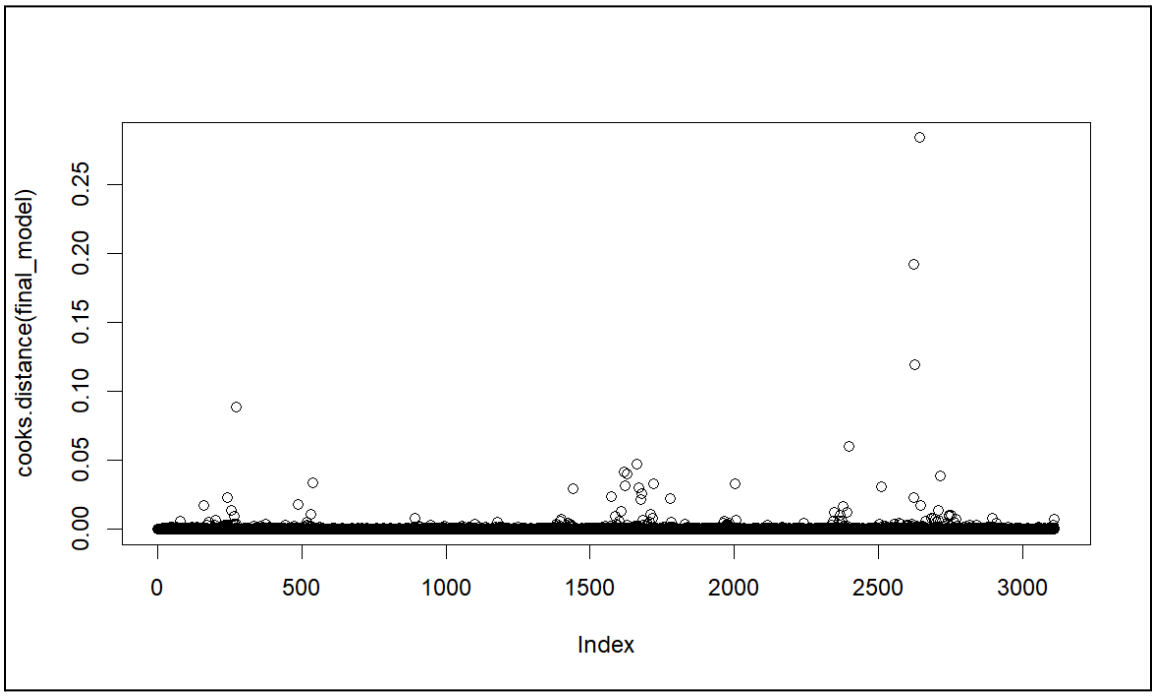

To check the normality of the distribution of the errors in the model, I generated a basic histogram (“hist(resid())” using my data object “model.” Looking at the resulting distribution curve, one can see that it is a fairly normal distribution with only slight tails on either side of the curve.

*Assumption 2:* Homoscedasticity of errors




To check the variance of errors across the model, I generated a fitted model plot which is reproduced above. Looking at the resulting plot, I would say that the model meets the criteria as the majority of the observations or plotted points are clustered together, demonstrating a level of consistency in the variance of the errors across the model.

*Assumption 3:* Absence of overly influential outliers



The last assumption check that I ran is calculating and plotting Cook's Distance values for the model. The resulting plot is dense and compact, signaling an absence of extreme outliers in general. While we can see a few examples of outliers, the vast majority of the plotted points are clustered together, almost forming a solid line.

Final Multicollinearity Check

Despite removing variables to attempt to address any multicollinearity, I ran the "vif()" function provided by the "car" library to check the variance inflation factor for my predictor variables. This test provides a value that reflects the likely multicollinearity of the variables within the overall model, and the higher the VIF the more of an issue that there may be for that specific variable. A table of the resulting VIF values for the model are below:

| Demographic Percentages | | | | Income |
|---|---|---|---|---|
| Asian | Black | Hispanic | NativeAm | Low |
| 1.456181 | 1.375246 | 1.187895 | 1.140626 | 1.411518 |
| Industry Percentages | | | | Partisanship Index |
| Agri Mining | Manufacturing | Retail | Arts Rec Food | Republican |
| 2.251486 | 1.663803 | 1.293698 | 1.432554 | 1.787698 |

Consulting the table, all of the variables' VIFs are between 1 and 3 which signals that they are not likely to be multicollinear and reflect a significant, independent relationship to the outcome variables.

<u>Linear Regression Results</u>

Reviewing all of the above plots and tables of the model indicate that it meets the necessary assumptions. With this confirmation, I continued analyzing and interpreting the final model.

| Formula: lm( audit_rate ~ ., data = dataset_filtered) | | | | |
|---|---|---|---|---|
| | Multilinear Regression Results | | | |
| Coefficient | Estimate | Std. Error | T Value | P- value |
| (Intercept) | 4.986 | 0.142 | 35.219 | <2e-16 |
| AsianPercent | 0.029 | 0.006 | 4.651 | 3.45.6 E-06 |
| BlackPercent | 0.034 | 0.001 | 32.507 | <2e-16 |

| | | | | |
|---|---|---|---|---|
| HispanicPercent | 0.026 | 0.001 | 16.126 | <2e-16 |
| NativeAmPercent | 0.030 | 0.002 | 13.755 | <2e-16 |
| LowerIncome | 4.770 | .1942 | 24.555 | <2e-16 |
| Agri_MiningPerc | -0.025 | 0.002 | -9.762 | <2e-16 |
| ManufacturingPerc | -.0145 | .002 | -6.203 | 6.28 E-10 |
| RetailPerc | 0.007 | .006 | 1.118 | 0.263 |
| Arts_Rec_FoodPerc | -0.025 | 0.004 | -5.912 | 3.76 E-09 |
| PARTISAN_INDEX_REP | 0.812 | 0.115 | 7.092 | 1.63 E-12 |
| F-statistic | 269.2 | | | |
| Adjusted R-Squared | .4631 | | | |

Looking at the results table above, there are immediate takeaways regarding the p-values for one of the variables in this model. The variable, RetailPerc, has a p-value significantly larger than .05 and so we cannot reject the null hypothesis for it. Since we cannot say that this variable has a statistically significant impact upon the audit rate, I will move on to interpreting the remaining variables, most of which have p-values significantly below .001.

Beginning with the “Lower Income” variable, we can see a very large potential effect upon the audit rate value, as the calculated coefficient value for “LowerIncome” is 4.9864. This value signifies that for each unit increase in the percentage of the population of the county that qualifies as lower income based upon my definition of the category, we would expect to see the audit rate increase by 4.9864 for that geographical region. Given that the audit rate ranges between 6 and 11, an increase of that magnitude is very significant. Looking at the low p-values and t-values for our demographic variables, one can see a statistically significant relationship between an increase in the percentage of the county’s households who identify as a racial or ethnic category other than white and the observed audit rate for that county. The largest connection is between Black-identified households and audit-rate where each percentage increase would likely result in an audit rate increase of .034. Similarly an increase in the percentage of the population of Asian-, Native American-, and Hispanic-identifying households corresponds to an expected increase in the audit rate of .02856, .03022, and .01653, respectively. Next, there are some trends emerging as well when one looks at the coefficients for the occupation variables. As the percentage of workforce engaged in the Agricultural/Mining, Manufacturing, and Arts/Recreation/Food Services industries increases, one would expect to see a decrease in the audit rates within that county by .0251, .0145, and .0252. This relationship may reflect the availability of higher payer jobs in those fields which corresponds with a lower observed audit rate for the county. The only occupational area whose increase in percentage of the workforce is associated with an increase in the county’s audit rate is that of Retail. Finally, the last remaining significant variable is that of the Partisan Index (Republican) which has a value of .8121, which is the highest coefficient value apart from the variable representing lower

income households. For each unit of increase in the Partisan Index (Republican), one would expect to observe a corresponding increase in the audit rate of that county by .8382.

To assess the goodness of fit of the model, I looked at the residual error, adjusted r-squared, and F- statistic values provided by the summary. With a residual standard error value of .7108, we would expect to observe actual audit rates that deviate by up to .7108 from our expected rate. Again, given that the audit rates range from 6 to 11, this variation is not completely minimal, however, it seems acceptable and indicative of a decent fit especially taking into account the multiple variables reflecting various different socioeconomic factors. Another indicator of the goodness of fit of the model is the F-statistic value of 269.2 with a p-value of almost 0, which indicates that the model with predictors provides a statistically better fit to the dataset than the model without any predictors. Finally, the r-squared value is .4649 meaning that approximately 46.5% of the variation within the dataset is explained by the model. The closer to 1 for the r squared value the better in terms of its explanatory power, however, .4645 reflects a strong fit for the model considering the complex set of variables within it.

## Initial Findings

After all the analysis, there are some interesting trends that emerge from the dataset which reflect previous inquiries as well as draw a few additional connections. Echoing Bloomquist's and the SIEPR's team's findings, there is a statistically significant connection between counties with higher percentages of racial and ethnic minority residents and higher audit rates. While their studies focused upon Black taxpayers, there is evidence that there is a racial disparity in those counties with the highest audit rates which corresponds to higher proportions of "Hispanic," "Asian," and "Native American" residents as well. In addition to this demographic relationship, the counties with higher proportions of lower median income households tended to have higher audit rates as well.

These results are all swirling around the fundamental core issue of IRS's treatment and enforcement surrounding the Earned Income Tax Credit program. Multiple teams of researchers have investigated and drawn the connection between higher audit rates specifically against recipients of the tax credit, whose eligibility is conditioned upon their lower income status. Due to this connection, my analysis may function as a sort of further fleshing out the recipients of the tax credit.

## Future Expansion

One route that could be pursued to provide a more nuanced understanding of these counties with higher audit rates would be to further break down these larger occupational sectors and refine the variables. I chose to use the total numbers for the individual sectors within the Census for my analysis to obtain a broader overview and capture general trends. While some connections surfaced, investigating smaller subsections of those industries might offer a more meaningful view of the experience and situation of the workers within that industry. If individual sections of the industry (perhaps, less skilled workers) tended to be associated with counties with these higher audit rates, it could point to additional linkages between lower income and higher audit rates.

Another way to expand this project would be to gather more recent data and apply these same analytical tests to the newer data sets. Bloomquist assembled and created the audit rates

data for the time period of 2012 through 2016. The IRS data exists to replicate his process and to extend the study up until 2020 if not further. To do so would require a fairly significant amount of data cleaning and re-organization, however, the resulting analysis would deepen the temporal element of the research. Whether a longer term analysis or comparing various four-year segments against one another, there are likely patterns and associations that would be revealed through the work.

## References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